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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2016-2017

CE2005/CZ2005 – OPERATING SYSTEMS

Nov/Dec 2016 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 7 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions carry equal marks.

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1. (a) Indicate whether the following statements are true or false. Justify your answers.

(i) All time-sharing systems are also real-time systems.
(2 marks)

(ii) Mutual exclusion can be achieved by disabling interrupts during the critical section.
(2 marks)

(iii) In Round-Robin scheduling, average waiting time is minimum when quantum size is equal to the largest CPU burst time among all processes.
(2 marks)

(iv) Deadlocks can occur even if the resource allocation graph does not have a cycle.
(2 marks)

(v) When a hardware interrupt arrives, the operating system will change the state of the running process from Running to Waiting.
(2 marks)

Note: Question No. 1 continues on Page 2

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(b) Consider the following set of processes to be scheduled in an operating system.

Table Q1

Process	Arrival Time	Total CPU Burst Time	I/O Burst Time
P0	0	3	2
P1	2	5	0
P2	4	2	2
P3	8	2	0

Process P0 first uses 1 unit of CPU burst time, followed by 2 units of I/O burst time and then completes the remaining 2 units of CPU burst time. Process P2 first uses 2 units of CPU burst time and then completes with 2 units of I/O burst time.

(i) Draw the schedule for the above processes under the following algorithms: a) shortest-job-first (non-preemptive) and b) round-robin with quantum size of 2 units.

(6 marks)

(ii) Compute the average waiting time under both algorithms.

(4 marks)

(c) Consider a semaphore S that uses blocking implementation. Suppose at the current time instant, a) 2 processes are holding S having successfully executed Wait(S) previously, and b) the blocked process list for S contains 4 process control blocks (PCBs).

(i) What is the value of S at the current time instant?

(1 mark)

(ii) Suppose the value of S changes to -1 after executing **Signal(S) five times and Wait(S) two times** in some order. What are the minimum and maximum intermediate values that S can have during these operations? Justify your answer by providing the sequence of operations that produce those values.

(2 marks)

(iii) What is the largest value that S can ever have? Justify your answer.

(2 marks)

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2. (a) Indicate whether the following statements are true or false. Justify your answers.

(i) System calls are executed in the kernel mode to support memory protection.

(2 marks)

(ii) When a context-switch happens from process P0 to process P1, register values are loaded from the PCB of P0 to the PCB of P1.

(2 marks)

(iii) When a process is in the Waiting state, it is waiting because another process is executing on the processor.

(2 marks)

(iv) The base and limit registers in a PCB are used for memory protection.

(2 marks)

(v) In shared memory based communication between processes, the shared global memory resides in kernel memory space.

(2 marks)

(b) Consider the following snapshot of a system's state, with four processes (P0, P1, P2 and P3) and three resource types (A, B and C). The current Allocation, Need and Available matrices are as shown in Table Q2.

Table Q2

		Available A B C 0 1 x
Process	Allocation A B C	Need A B C
P0	2 1 1	0 1 0
P1	1 1 0	2 1 2
P2	1 1 1	2 0 1
P3	1 1 1	4 1 0

(i) Compute the minimum value for x in the Available matrix of Table Q2, so that the above system state is safe. Justify your answer.

(4 marks)

Note: Question No. 2 continues on Page 4

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(ii) Suppose process P1 generates the resource request (A=0 B=1 C=2) next. Is there a value of x such that this request can be granted and the resulting system state is safe?

(3 marks)

(c) Alice is playing a multiplayer game that involves pairing apples with oranges. The game produces apples and oranges in separate baskets at random time instants. For Alice to get points, she must **pair exactly 1 apple with 2 oranges** by picking them from the respective baskets.

Each basket is protected by a semaphore; A for the apple basket and O for the orange basket. To access a basket, Alice must first acquire the corresponding semaphore. A picked fruit will attract penalty points until it is paired. The penalty only depends on the time duration for which the fruit is held without being paired, and in particular does not depend on the number or type of fruit.

(i) Complete the code in Figure Q2 to help Alice to play this game. You should only use the operations Wait(A), Wait(O), Signal(A) and Signal(O) in the boxes provided. You may use any number of these operations (including zero) in each of the boxes. **Your solution must be deadlock-free even in the presence of other players who may be executing a different code**.

```
[ 1 ]
// Pick 1 apple
[ 2 ]
If (atleast 2 oranges in basket) {
  // Pick 2 oranges and pair with apple.
[ 3 ]
} else {
[ 4 ]
// Drop 1 apple back in basket
[ 5 ]
}
```

Figure Q2

(5 marks)

(ii) Does the above solution minimize the penalty points for Alice? Justify your answer.

(3 marks)

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3. (a) Indicate whether the following statements about memory management are true or false. Justify your answers.

(i) Memory allocation using segmentation approach suffers from internal fragmentation.

(2 marks)

(ii) Translation Look-aside Buffer (TLB) is used to reduce page faults.

(2 marks)

(iii) With demand paging, address binding must be performed during load time.

(2 marks)

(iv) As the number of available page frames increases, the number of page faults will always decrease no matter which page replacement algorithm is used.

(2 marks)

(v) Thrashing situation can be temporarily alleviated by swapping out some processes.

(2 marks)

(vi) Both local and global page replacement policies can be used under the variable frame allocation scheme.

(2 marks)

(b) Consider a two-level paging scheme where 32-bit logical addresses are partitioned as follows:

9 bits	10 bits	13 bits

(i) What is the page size?

(2 marks)

(ii) How many entries are there in the outer level page table?

(2 marks)

(iii) Draw a diagram to depict how a logical address is mapped to its physical address using this two-level paging scheme.

(3 marks)

(iv) If single-level paging scheme is used, how many entries are there in the page table?

(2 marks)

Note: Question No. 3 continues on Page 6

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(c) Discuss how the following programming styles may affect the performance in a demand paging system:

- Use of arrays
- Use of linked lists

(4 marks)

4. (a) Indicate whether the following statements about file system and I/O are true or false. Justify your answers.

(i) Using tree-structured directories, different files can have the same name under different directories.

(2 marks)

(ii) Assuming the protection bits of a file are "r - x r - x - - x", all the users in this Unix-based system are able to read this file.

(2 marks)

(iii) Caching is used to store data temporarily in memory when they are transferred between devices with different transfer sizes.

(2 marks)

(iv) A program reading a file sequentially may result in a wide swing of disk head if contiguous allocation method is used for allocating storage space for files.

(2 marks)

(v) Asynchronous I/O is typically used in the implementation of Graphical User Interface (GUI).

(2 marks)

(vi) Mirror of strips has better reliability than strips of mirrors.

(2 marks)

Note: Question No. 4 continues on Page 7

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(b) Assume that a file system uses Unix-like inodes with 12 direct pointers and one single-indirect pointer for file allocation and that a directory file can fit in a single data block. Initially, all files and inodes are in disk.

(i) If the current working directory is "/spell/mail" which is already in memory, how many disk read operations are required to open file "/spell/mail/ptr/first"?

(3 marks)

(ii) After the file is opened, to read data from its 10th to 13th block, how many disk read operations are further required?

(3 marks)

(c) Assume that a disk drive has 200 cylinders, numbered 0 to 199. The disk head is currently at cylinder 160 and the following I/O requests are in the disk queue: 30, 60, 150, 180, and 190.

(i) To serve these requests in the queue, which disk scheduling algorithm, SSTF or SCAN, will result in less head movement? For the SCAN algorithm, assume disk head is currently moving towards cylinder 199.

(4 marks)

(ii) Discuss the advantage of SCAN over SSTF algorithm.

(3 marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.